A Wavelet-Augmented Dual-Branch Position-Embedding Mamba Network for Hyperspectral Image Change Detection

Chen Ding, *Member, IEEE*, Xiaofeng Hao, Sirui Zheng, Yizhou Dong, Wenqiang Hua, *Member, IEEE*, Wei Wei, *Senior Member, IEEE*, Lei Zhang, *Member, IEEE*, and Yanning Zhang, *Fellow, IEEE*

***Abstract*—Convolutional neural networks (CNNs) have been widely used in the field of hyperspectral image change detection (HSI-CD) due to their excellent feature extraction capabilities. However, traditional convolution methods usually rely on a limited number of convolution kernels when extracting local features, and the local information only covers a small range of details, making it difficult to flexibly extract local changes at different levels in the image. In addition, CNNs have certain limitations in modeling global dependencies. In recent years, state-space model (SSM)-based methods (such as Mamba) have effectively solved the quadratic complexity problem of the transformer in capturing long-distance global dependencies. However, due to the processing method of expanding images into sequences, the spatial location structure of the image is destroyed, the spatial correlation between adjacent pixels is weakened, and spatial location information is easily lost, resulting in poor performance on complex hyperspectral datasets. To address these issues, we propose a wavelet-augmented dual-branch position-embedding Mamba network (WDP-Mamba) for hyperspectral image change detection (CD). This network combines the complementary advantages of Mamba and wavelet convolution (WTconv) in global feature extraction and local edge feature extraction. Specifically, our method includes the following key innovations: 1) a novel state-space model, the adaptive position residual state space block (APRSSB) is designed, embedding learnable positional encodings within the Mamba framework. Through a dual-temporal spatial–spectral scoring mechanism, the model dynamically preserves the topological relationships of pixel positions, enabling global dependency modeling with linear complexity and 2) a multilevel frequency-domain aware (MLFA) module is proposed to fully extract representative local features by decomposing information into frequency components of different scales and analyzing signals locally in multiple time and frequency windows. Extensive experiments on three hyperspectral datasets show that WDP-Mamba outperforms existing methods in both qualitative and quantitative assessments. Compared with other comparative models, WDP-Mamba achieves significant improvements in multiple performance metrics with a lower number of parameters, demonstrating its superior performance in the HSI-CD task.**

***Index Terms*—Change detection (CD), Mamba, spatial-spectrum position relationship, wavelet transform (WT).**

I. Introduction

HYPERSPECTRAL as a imaging technology, as a cutting-edge method in the field of remote sensing, is rapidly becoming a core tool for Earth observation and change detection (CD) research due to its unique hyperspectral resolution and rich spectral information [1], [2], [3]. Unlike traditional multispectral images [4], hyperspectral images, obtained through detailed and continuous spectral band data, not only enable the analysis of the spectral characteristics of surface objects but also allow for more precise land classification and detection of change areas [5]. Hyperspectral CD (HSI-CD) is an important technique for identifying changes in land cover by comparing hyperspectral images of the same region captured at different times. This technique has been widely applied in fields such as ecological monitoring [6], natural disaster assessment [7], and land use change analysis [8], playing a pivotal role, particularly in key applications like precision agriculture and environmental protection [9], [10]. However, the high dimensionality of hyperspectral images, which leads to spectral redundancy and high computational costs, presents a significant challenge for HSI-CD.

Traditional hyperspectral CD methods can be broadly categorized into three types: algebraic-based methods, pixel classification-based methods, and image transformation-based methods. Algebraic-based methods, such as change vector analysis (CVA) [11], calculate the spectral differences of a pixel at two distinct time points to derive a change vector. Liu et al. [12] proposed S2CVA, which introduces a band-weighting strategy that assigns different weights to various spectral bands, enhancing the influence of those bands relevant to the change and reducing the impact of bands unrelated to the change, such as noise bands or seasonal variation bands.

Received 31 March 2025; revised 16 July 2025; accepted 18 August 2025. Date of publication 25 August 2025; date of current version 11 September 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62472350, Grant 62472359, and Grant 62372379; and in part by the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing Foundation under Grant TJ-04-23-04. *(Corresponding author: Lei Zhang.)*

Chen Ding, Xiaofeng Hao, Sirui Zheng, Yizhou Dong, and Wenqiang Hua are with the School of Computer Science and Technology, Shaanxi Key Laboratory of Network Data Analysis and Intelligent Processing, and Xi'an Key Laboratory of Big Data and Intelligent Computing, Xi'an University of Posts and Telecommunications, Xi'an 710121, China (e-mail: dingchen@xupt.edu.cn; hxf@stu.xupt.edu.cn; ZhengSirui@stu.xupt.edu.cn; dyzhuo@xupt.edu.cn; huawenqiang@xupt.edu.cn).

Wei Wei, Lei Zhang, and Yanning Zhang are with Shaanxi Provincial Key Laboratory of Speech and Image Information Processing and the National Engineering Laboratory for Integrated Aerospace-Ground-Ocean Big Data Application Technology, School of Computer Science, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: weiweinwpu@nwpu.edu.cn; nwpuzhanglei@nwpu.edu.cn; ynzhang@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3602225

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

However, algebraic-based methods are susceptible to noise and are unable to handle redundant information effectively. Pixel classification-based methods are generally viewed as binary classification tasks. Nemmour and Chibani [13] proposed a land cover CD approach based on multiple support vector machines (SVMs), combining several SVM classifiers, each focusing on remote sensing images from different time periods, thereby enhancing the model's ability to detect complex changes. Image transformation-based methods typically rely on transforming hyperspectral images from the original spatial domain to other feature spaces. These include principal component analysis (PCA [14]), which employs linear transformations to project data into a new feature space, maximizing the variance of the data to emphasize change features. Bruce et al. [15] introduced a multiresolution signal processing method, using discrete wavelets to project the signal onto a wavelet basis, thereby separating the image's detail and approximation information. In addition, methods such as multivariate CD (MAD) [16] and slow feature analysis (SFA) [17] map the original spatial domain to other feature spaces, effectively reducing the dimensionality of the data and highlighting more prominent change features.

With the development of deep learning networks, convolutional neural network (CNN)-based methods have achieved remarkable success in HSI-CD and have gradually become one of the mainstream approaches for CD. CNNs excel at extracting multilevel feature representations, surpassing traditional handcrafted feature extraction algorithms. Through end-to-end CNN methods, change areas can be effectively detected. For instance, Chen et al. [18] proposed a 2D-CNN network, which extracts deep features from hyperspectral images (HSIs) through multiple convolutional and pooling layers, enhancing the nonlinearity, discriminability, and invariance of the features. Wang et al. [19] introduced an end-to-end 2-D CNN model called GTETNET, which effectively integrates multisource information by introducing a mixed affinity matrix. Li et al. [20] proposed an unsupervised fully convolutional network based on deep noise modeling, addressing the interference of noise in hyperspectral data for CD, demonstrating its effectiveness under low signal-to-noise ratio conditions. In addition, Hou et al. [21] proposed the SALA method, which utilizes the spectral angle weighted information from the target pixel and its surrounding eight neighboring pixels, further enhancing the local differences between pixels. This method preserves the local characteristics of the target pixel while effectively suppressing background noise. In addition to extracting spatial local information, spectral information with rich channel dimensions has also been fully utilized. Wang et al. [22] proposed a spectral–spatial CD method based on a Siamese network, which jointly learns spectral and spatial features to enhance the detection of complex surface changes. Hu et al. [23] introduced the self-supervised hyperspectral spatial–spectral feature understanding network (HyperNet), which extracts discriminative spatial and spectral information separately. Although these methods have made significant progress in extracting spatial and spectral information, they are limited by their inability to fully capture the features of complex scenes due to their reliance on local information. As a result, researchers have subsequently introduced multiscale feature representation methods to better meet the demands of CD. For example, Qu et al. [24] proposed a multilevel encoder–decoder attention network (ML-EDAN), which effectively extracts and transfers spatial–spectral information between low-level and high-level features by incorporating contextual information to guide attention, thereby achieving multiscale feature fusion. Feng et al. [25] proposed a multiscale fusion network (MsFNet), which extracts multiscale features from deep feature maps using convolution kernels of varying sizes and fuses information at different scales through layer-wise upsampling and feature stacking operations. The aforementioned CNNs and their variants effectively leverage the spatial–spectral features of hyperspectral data, greatly enhancing performance in the field of CD. Although many studies [26], [27], [28] have achieved effective feature extraction at different scales through the design of convolutional kernels, these methods are limited by the fixity of the local receptive field, which limits their ability to adaptively capture spatial–spectral features at different levels. This results in an attenuation of the response to high-frequency details in complex scenes, which weakens the model's ability to recognize subtle changes.

Recently, transformer [29] have demonstrated remarkable performance in natural language processing [30] and have been successfully adapted to vision tasks and remote sensing applications [31]. A transformer can simultaneously attend to spatial and spectral information on a global scale, enabling more precise characterization of land cover differences and extraction of change regions in CD tasks [32]. For instance, Wang et al. [33] proposed an end-to-end transformer that jointly leverages spectral, spatial, and temporal information. This approach enhances fine-grained spectral representations through positional embeddings and extracts spatial texture information from local image patches, thereby modeling nonlocal relationships between pixels. Song et al. [34] introduced a cross-temporal interaction symmetric attention network (CSANet) to fuse differential features embedded across different time phases. In addition, researchers have proposed hybrid methods combining CNNs and transformers. For example, Chen et al. [35] developed a triple-path local-to-global network (TL2GH2T), which employs three parallel pathways to learn spatial, spectral, and spatial–spectral features separately. Nevertheless, the self-attention mechanism inherent in transformer architectures exhibits quadratic complexity with respect to sequence length, thereby imposing substantial constraints on their efficacy in processing the inherent high-dimensional nature of hyperspectral data cubes. This limitation manifests as prohibitive computational and memory overhead, particularly when applied to large-scale remote sensing imagery with expansive spatial–spectral dimensions.

To overcome the limitations of the transformer, Gu and Dao [36] recently proposed the Mamba network based on state space models (SSMs), which enables global modeling with linear complexity. It has achieved remarkable performance in natural language processing tasks and is regarded as an effective alternative to the transformer. Inspired by the success of Mamba, Zhu et al. [37] introduced vision Mamba, pioneer-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

ing the application of Mamba in visual tasks. Subsequently, Mamba models have been widely applied to visual tasks [37], [38], [39] including numerous methods in the medical imaging processing field, such as Mamba-Unet [40], VM-Unet [41], U-Mamba [42], LightM-Unet [43], and SegMamba [44]. Mamba-based methods have also demonstrated significant capabilities in remote sensing tasks [45], [46], [47], [48], [49], [50], [51], [52]. Due to the selective scanning mechanism of SSM, Mamba can achieve a larger receptive field in specified directions, and researchers have made several improvements to this scanning mechanism. Liu et al. [53] proposed forward and backward scanning to model flattened visual features in both directions. Zhao et al. [54] introduced an omnidirectional scanning mechanism that selectively scans from horizontal, vertical, diagonal, and anti-diagonal directions, thus enabling global background modeling of remote sensing images. Although these improvements have led to significant success for Mamba in natural image processing, the architecture of the SSM causes Mamba to flatten 2-D images into 1-D sequences, forming a series of tokens as input. This process disrupts the original spatial structure of the image, resulting in the loss of contextual information. Merely modifying the scanning mechanism cannot preserve the original spatial positional relationships in the image. On the contrary, it may lead to the incorrect classification of pixels with similar spatial–spectral features located in different positions.

To address the aforementioned limitations, we propose a Mamba-based network specifically tailored for hyperspectral image CD (HSI-CD), referred to as WDP-Mamba. The network framework is illustrated in Fig. 1. To the best of our knowledge, this represents the preliminary exploration of Mamba in the context of HSI-CD. In summary, the key contributions of our work are outlined as follows.

1) To address the limitations of CNNs in extracting local variation features across different levels within an image and the computational overheads associated with the transformer in capturing long-range dependencies, we introduce WTconv and Mamba, integrating both into a dual-branch architecture to construct a novel network model, WDP-Mamba. This design leverages Mamba's linear complexity global modeling capability to capture long-range spectral–spatial information, while WTconv performs multiscale wavelet decomposition to extract frequency-domain local features. The cross-branch feature fusion mechanism ensures complementary enhancement between global contextual information and local frequency-domain features, thereby enabling robust representation learning for subtle variations.
2) To mitigate the spatial structure destruction, which is induced by Mamba's sequence flattening that destroys the spatial location structure and weakens the correlation between adjacent pixels, we propose the APRSSB. This module introduces dual-temporal spatial–spectral scoring that dynamically recalibrates region-wise importance weights through learnable positional embeddings. By preserving interpixel topological relationships and spectral band correlations, APRSSB achieves precise localization of change boundaries while maintaining computational efficiency.
3) To address the limitation of traditional convolutional methods, which capture only localized details within a narrow scope and struggle to adaptively extract local variations across multiple levels of an image, we propose a multilevel frequency-domain aware (MLFA) module based on WTconv. This approach decomposes the image information into multiple frequency components through wavelet transforms (WTs), enabling localized analysis across various frequency windows and scales, thereby facilitating the perception of representative local features at multiple hierarchical levels.

The article is organized as follows. Section II covers SSM preliminaries. Section III details the WDP-Mamba network's framework and core components. Section IV presents the experimental datasets, configurations, ablation studies, and analyses. Section V examines the effects of various convolution methods and settings. Section VI summarizes the work and discusses future directions.

II. Preliminaries

The SSM originally stems from the classical Kalman filter model, with its core concept being the establishment of a relationship between inputs and outputs in a time series through hidden states $h(t)$. The efficiency and performance of SSM in modeling long sequences have made it a prominent focus of research in recent years. The continuous form of the model is typically defined as follows:

$$h'(t) = Ah(t) + Bx(t) \tag{1}$$

$$y(t) = \mathrm{Ch}(t) \tag{2}$$

where $A \in \mathrm{R}^{N\times N}$ is the state transition matrix, and $B \in R^{N\times 1}$ and $C \in R^{1\times N}$ are the projection matrices for the input and output, respectively.

To transform a continuous-time system into its discrete-time counterpart, the zero-order hold (ZOH) method is frequently employed for parameter discretization. The resulting discretized system can be represented as

$$A = e^{\Delta \mathrm{A}} \tag{3}$$

$$B = (\Delta A)^{-1}\left(e^{\Delta A} - I\right) \cdot \Delta B \tag{4}$$

in this context, Δ denotes the time step, and I represents the identity matrix. Through the process of discretization, the state update equation is transformed as follows:

$$h_{\mathrm{t}} = Ah_{t-1} + Bx_t \tag{5}$$

$$y_t = \mathrm{Ch}_t. \tag{6}$$

Furthermore, the output computation can be efficiently achieved through convolutional operations. The expression for the convolutional kernel K is given by

$$K = \left(C\overline{B}, C\overline{AB}, C\overline{A}^2\overline{B}, \ldots, C\overline{A}^{M-1}\overline{B}\right). \tag{7}$$

Consequently, the computation of the output y can be expressed as

$$y = x * K \tag{8}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

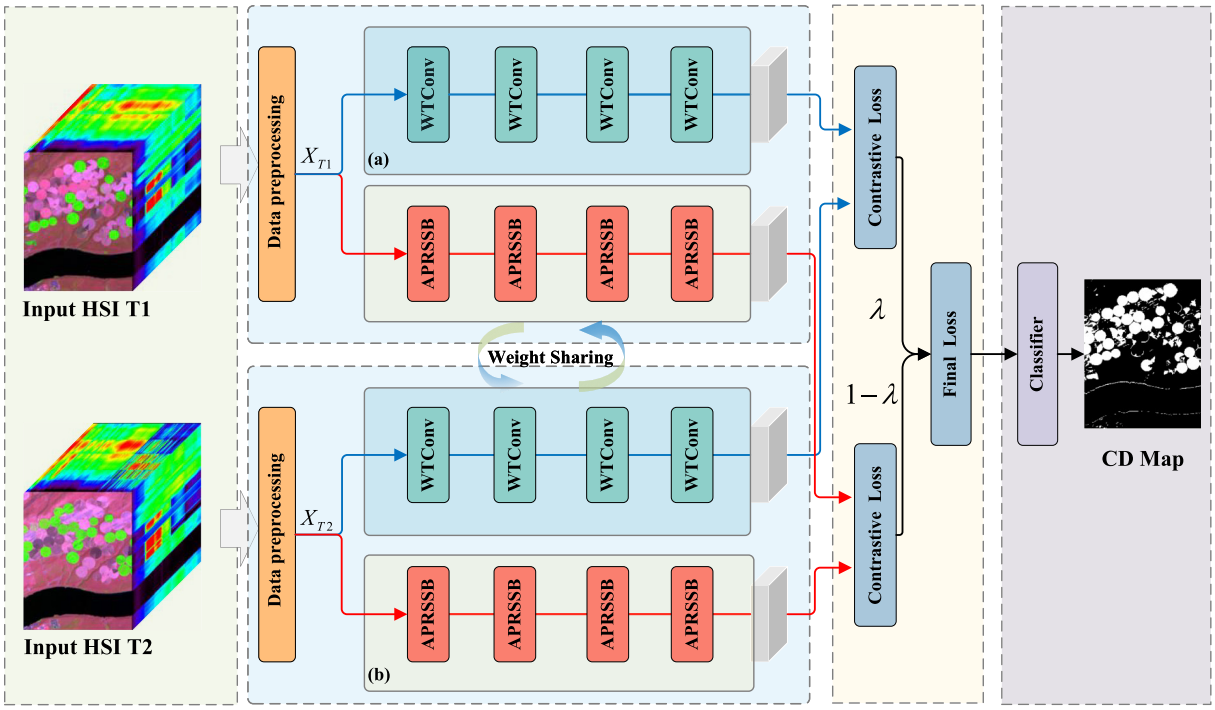


Fig. 1. Overall architecture of the WDP-Mamba that we proposed is composed of two parallel and complementary branches. The lower global modeling branch leverages a state space mechanism to capture long-range dependencies, while the upper local representation branch employs WTconv to extract multilevel local features. The global branch utilizes sequence flattening and positional embedding to preserve spatial semantic information, whereas the local branch maintains local details through multifrequency decomposition. By fusing global and local information, the model ultimately generates accurate CD maps through a classifier. (a) Multilevel frequency-domain aware. (b) Adaptive position residual state-space group.

where * denotes the convolution operation. Building upon the foundation of discretization, the SSM model further incorporates data-dependent mechanisms, enabling them to adapt efficiently to hardware-optimized computational environments. In contrast to traditional self-attention mechanisms, SSMs exhibit a near-linear computational complexity, making them particularly well-suited for processing long-sequence data. These characteristics endow SSMs with significant potential for applications in vision-related tasks.

Directly applying SSMs to hyperspectral image CD poses challenges due to the need to preserve spatial–spectral structure. Tokenization and flattening used in SSMs often disrupt spatial continuity, hindering the precise localization of changes. Thus, a tailored framework is needed–one that maintains spatial coherence while leveraging the global modeling strengths of SSMs. This motivates the integration of complementary strategies for balanced global and spatial representation learning, as detailed in Section III.

III. Methods

This study presents a wavelet-augmented dual-branch position-embedding Mamba network (WDP-Mamba), as illustrated in Fig. 1, designed to accurately capture the differential information between bi-temporal hyperspectral images. Specifically, we introduce two pivotal modules: the MLFA module and the APRSSB, which are independently integrated into the two branches of the Siamese network for deep extraction of both local and global features of the HSI. The MLFA module employs WTconv at multiple levels to decompose the information into frequency components across various scales, facilitating localized analysis within different frequency windows and scales, thereby effectively addressing the spatial–spectral nonstationary signals in hyperspectral data. The APRSSB utilizes an interactive structure to score the positional information across both the channel and spatial dimensions. To mitigate the risk of over-relying on any single dimension and thus disrupting the balance of positional representations, we design learnable dynamic parameters for more efficient allocation of positional weights. The output features are subsequently classified via a fully connected layer to generate the final detection results. Furthermore, by sharing parameters within the Siamese network, we effectively reduce computational costs, enabling the dual-branch network to maintain consistent feature extraction capabilities, thereby more accurately capturing the discrepancies between bi-temporal images.

A. Data Preprocessing

In this study, we treat hyperspectral images as 3-D data cubes, denoted as $O \in R^{H\times W\times C}$, where H and W represent the image height and width, respectively, and C denotes the spectral dimension. To accurately capture the local spatial–spectral information, we extract 3-D patches at each pixel location. Specifically, a 5×5 neighborhood window is selected; for pixels at the image edges that do not satisfy the window size, the image is padded by $(s-1)/2$ pixels in all directions (with $s = 5$), ensuring that the neighborhood data blocks for all pixels maintain a uniform size of $5 \times 5 \times C$. After completing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

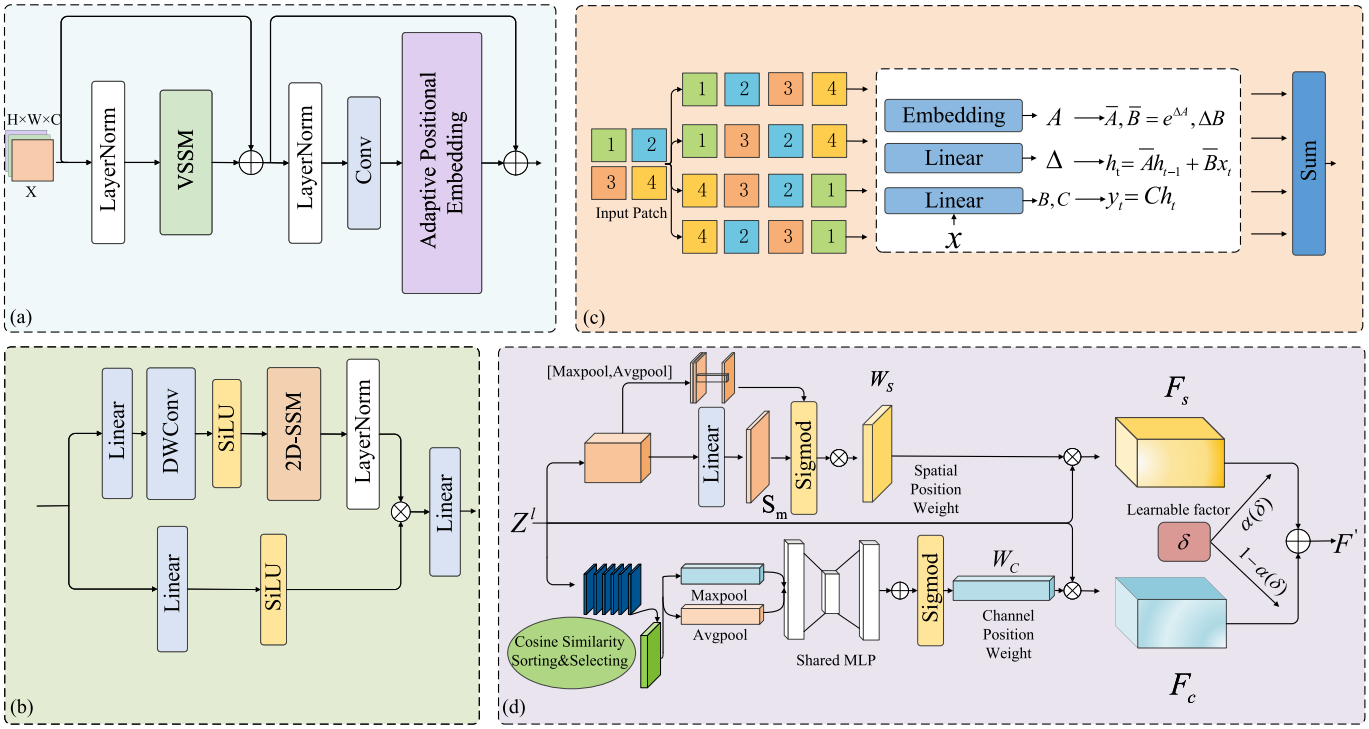


Fig. 2. Network structure of the adaptive positional residual state space block (APR-SSB) is designed to efficiently model long-range dependencies using a state space mechanism while preserving spatial structures that are often disrupted by sequence flattening. The adaptive positional embedding (APE) unit embeds the lost spatial-channel positional information and introduces learnable factors to dynamically select the most informative positional cues. This enables the model to maintain spatial topological consistency during the global modeling process. (a) APRSSB. (b) Vision state space module (VSSM). (c) 2-D-selective scan module (2D-SSM). (d) Adaptive positional embedding.

the spatial preprocessing, the same operation is applied to hyperspectral images from two different time points ($T1$ and $T2$), ensuring that the patches from both time periods are aligned in terms of both spatial and spectral dimensions. It is noteworthy that, based on the dataset characteristics, invalid pixel patches are filtered out to remove anomalous samples, thereby reducing interference during subsequent training.

Finally, the resulting patches are randomly divided into training and testing sets according to a predefined ratio, which ensures a diverse training dataset while facilitating an objective evaluation of the model's generalization performance under different scenarios. Consequently, the collections of patches from the two time points are denoted as $X_{T1} \in R^{N\times5\times5\times C}$ and $X_{T2} \in R^{N\times5\times5\times C}$, where N represents the total number of patches and each patch is represented as $X \in R^{5\times5\times C}$. These patches serve as the fundamental input for the subsequent CD tasks.

B. Adaptive Positional Residual State-Space Block

The introduction of the variant Mamba model offers an efficient mechanism for sequence modeling. Its 2D-SSM scans the hyperspectral data cube along multiple directions, such as horizontal, vertical, diagonal, and anti-diagonal, and unfolds it into sequential representations. This multidirectional scanning strategy ensures that each pixel establishes dependencies with its spatial neighbors through at least one directional path. By leveraging the structured recurrent modeling capability of SSMs, 2D-SSM enables global spatial modeling in hyperspectral images without relying on computationally expensive self-attention mechanisms. While this sequential modeling enhances the representation of global contextual information, it inevitably introduces a drawback: flattening the spatial structure into 1-D sequences can cause the loss of positional cues and topological relationships inherent in the original image. This makes it difficult for the model to preserve the spatial topology of the image. To enable Mamba to perform global mapping more efficiently and better adapt to CD tasks, we propose an adaptive positional residual state-space block to retain spatial positional relationships. The details of the adaptive positional residual state-space block are illustrated in Fig. 2.

The adaptive positional residual state-space block is constructed based on the VSS block. Specifically, the input image patches $X \in R^{H\times W\times C}$, where $H\times W\times C$ represent the HSI data cube with preprocessed dimensions of $5\times5\times C$, respectively. It is first processed by a LayerNorm layer, then fed into the VSS block to model long-range dependencies, while a skip connection is employed to enhance gradient flow and feature retention. Finally, the output representation of the state update is obtained as follows:

$$Z^l = \mathrm{VSSM}\left(\mathrm{LN}\left(X\right)\right) + X. \tag{9}$$

Furthermore, due to the flattening strategy employed by the VSS block, which processes the feature map into a 1-D token sequence, it encounters challenges in modeling spatial positional relationships. To address this limitation, we introduce an adaptive positional embedding after the VSS block to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

restore positional correlations. Specifically, we first apply a LayerNorm layer to normalize Z^l. Subsequently, to preserve positional information, the normalized output is fed into the adaptive positional embedding, which infers two independent weighted feature maps representing channel-wise and spatial-wise weights, respectively. For spatial weighting, a linear regression layer is utilized to generate a saliency map S_m, where $S_m(i, j) \in [0, 1]$ denotes the probability of whether the pixel location (i, j) is likely to undergo changes

$$S_m = \sigma\left(f_c\ \left(Z^l\right)\right) \tag{10}$$

where σ represents the sigmoid activation function. Note that f_c refers to the fully connected layer, which is a linear regression layer. Then, by combining global average pooling and max pooling operations, the spatial attention map M_s is generated

$$M_s = \sigma\left(\mathrm{Conv}^{1\times 1}\left(\left[\mathrm{AvgPool}^{(Z^l)}; \mathrm{MaxPool}^{(Z^l)}\right]\right)\right) \tag{11}$$

where σ denotes the sigmoid activation function, $\mathrm{Conv}_{1\times 1}$ represents a convolutional operation with a filter size of 1×1, and AvgPool and MaxPool refer to global average pooling and global max pooling operations applied along the channel dimension, respectively. $M_s \in \mathbb{R}^{H\times W}$ is the attention map generated from the spatial relationships of the features. Finally, the spatial positional weights are $W_s \in R^{H\times W\times 1}$ obtained by fusing the saliency map S_m and the spatial attention map M_s

$$W_s = S_m \otimes M_s \tag{12}$$

the feature map is then weighted to obtain the reweighted featureF_s

$$F_s = W_s \otimes Z^l. \tag{13}$$

Since changes in chemical composition and other factors in hyperspectral images can lead to significant differences in spectral information, it is essential to localize channel-wise positional information and identify which regions along the channel dimension are more likely to contribute to changes. Similar to the spatial positional embedding, for the channel weighting component, we compute the cosine similarity along the spectral dimension for each pixel between the deep feature maps $Z^l_{T1} \in R^{H\times W\times C}$ at time $t1$ and $Z^l_{T2} \in R^{H\times W\times C}$ at time $t2$, in order to obtain the change response values for specific channels. Let each pixel location be (x, y), and its corresponding spectral feature vectors be $Z^l_{T1}(x, y) \in R^{1\times 1\times C}$ and $Z^l_{T2}(x, y) \in R^{1\times 1\times C}$. The cosine similarity between these two spectral vectors is calculated as follows:

$$S\,(x, y) = \frac{Z^l_{T1}\,(x, y) \cdot Z^l_{T2}\,(x, y)}{\left\|Z^l_{T1}\,(x, y)\right\| \left\|Z^l_{T2}\,(x, y)\right\|} \tag{14}$$

where $\cdot$ denotes the vector dot product, and $\|\cdot\|$ represents the vector norm. Subsequently, for each pixel, the response value $S(x, y)$ along the spectral dimension is computed. The response values for each feature channel are then sorted to obtain a spectral response sequence ranked from high to low. The top K values are selected to capture the most significant change features $\{S^1(x, y), S^2(x, y), \ldots, S^K(x, y)\}$ in the spectral channels, where $S^1(x, y) \geq S^2(x, y) \geq \ldots \geq S^K(x, y)$. The average of the selected K response values is computed to obtain the final activation value for the position (x, y)

$$V\,(x, y) = \frac{1}{K}\sum_{k=1}^{K} S^k\,(x, y) \tag{15}$$

the activation values $V(x, y)$ generated for each pixel together form a 2-D feature map $V \in R^{H\times W}$. We use average pooling and maximum pooling to integrate the channel information of the feature map, thus generating two spatial context descriptions V_a (average pooled features) and V_m (maximum pooled features), respectively, which are subsequently fed into a shared perceptron to generate the final channel position weights $W_c \in R^{1\times 1\times C}$

$$W_c = \sigma\left(\mathrm{MLP}\left(\mathbf{V_a}\right)\right) + \sigma\left(\mathrm{MLP}\left(\mathbf{V_m}\right)\right). \tag{16}$$

Note that the weights of the MLP are shared for both inputs, with the first layer utilizing the ReLU activation function. The feature map is then weighted to obtain the reweighted feature F_c

$$F_c = W_c \otimes Z^l. \tag{17}$$

Finally, to selectively emphasize positional information features and suppress less significant positional features, enabling more precise weight allocation, the channel-wise weighted feature F_c and the spatial-wise weighted feature F_s are combined using an adaptive fusion strategy to generate the final fused weight matrix F'. Specifically, when assigning weights to each channel's position, a learnable balancing factor δ is introduced to dynamically balance the channel and spatial weights, resulting in the final fused feature map $F' \in R^{H\times W\times C}$. The formula for calculating the fusion weights is as follows:

$$F' = \alpha\left(\alpha\left(\delta\right) \cdot \alpha\left(F_c \otimes F_s\right) + \left(1 - \alpha\left(\delta\right)\right) \cdot \alpha\left(F_c\right)\right) \tag{18}$$

where $F_c \otimes F_s$ denotes the positional interaction enhancement term, which explicitly constructs joint dependencies between spatial coordinates and spectral order through element-wise multiplication. When spectral sequences exhibit abrupt changes at specific spatial locations, this interaction term enhances the weight allocation toward position-sensitive regions. In contrast, F_c serves as the spectral positional reference term, ensuring the stability of spectral ordering without being affected by spatial correlations. To dynamically balance these two paths, we introduce a learnable parameter δ, which is normalized via a sigmoid function to produce a fusion coefficient $\alpha(\delta)$. This coefficient modulates the strength of interaction enhancement: $\alpha(\delta)$ emphasizes joint modeling in regions with strong spatial–spectral co-variation, while $1-\alpha(\delta)$ preserves the independence of spectral ordering, preventing confusion caused by excessive interaction. In this way, the model achieves adaptive enhancement by selectively integrating spatial and spectral positional information.

C. Multilevel Frequency-Aware Module

In order to more effectively capture the multilevel local details and frequency characteristics within hyperspectral data, we introduce WTconv. Hyperspectral data, as a 3-D discrete

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 WTConv Layer

Input: $X \in \mathbb{R}^{H\times W\times C}$
Output: $Z^{(0)}$
$X_{\mathrm{LL}}^{(0)} \leftarrow X$
$Y_{\mathrm{LL}}^{(0)} \leftarrow \mathrm{Conv}(W^{(0)}, X_{\mathrm{LL}}^{(0)})$
for $i = 1$**to**ℓ **do**
  $X_{\mathrm{LL}}^{(i)}, X_{\mathrm{LH}}^{(i)}, X_{\mathrm{HL}}^{(i)}, X_{\mathrm{HH}}^{(i)} \leftarrow WT(X_{\mathrm{LL}}^{(i-1)})$
  $Y_{\mathrm{LL}}^{(i)}, Y_{\mathrm{LH}}^{(i)}, Y_{\mathrm{HL}}^{(i)}, Y_{\mathrm{HH}}^{(i)} \leftarrow \mathrm{Conv}(W^{(i)}, (X_{\mathrm{LL}}^{(i)}, X_{\mathrm{LH}}^{(i)}, X_{\mathrm{HL}}^{(i)}, X_{\mathrm{HH}}^{(i)}))$
end for
$Z^{(\ell+1)} \leftarrow 0$
for $i = 1$**to**ℓ **do**
  $Z^{(i)} \leftarrow \mathrm{IWT}(Y_{\mathrm{LL}}^{(i)} + Z^{(i+1)}, Y_{\mathrm{LH}}^{(i)}, Y_{\mathrm{HL}}^{(i)}, Y_{\mathrm{HH}}^{(i)})$
end for
$Z^{(0)} \leftarrow Y_{\mathrm{LL}}^{(0)} + Z^{(1)}$
return $Z^{(0)}$

nonstationary signal, encompasses rich spatial and spectral information while spanning a wide range of frequencies. Traditional convolutional operations, when applied to such data, often struggle to explicitly separate high-frequency details from low-frequency global information. Moreover, unlike traditional Fourier transforms, which provide only global frequency-domain information and fail to localize time-varying features, WTs, through localized analysis, can accurately capture frequency characteristics that vary over time or space, thus offering a more robust solution for handling signals with time-varying properties. Haar WTs, in particular, facilitate multiresolution decomposition of hyperspectral signals, enabling the extraction of local features across different frequency windows and scales. Building on this foundation, we employ the WTConv layer to extract frequency-domain and multiscale information. The specific structure is illustrated in Fig. 3.

WTConv transforms spatial features into four frequency-domain components using WTs: one low-frequency component and three high-frequency components corresponding to horizontal, vertical, and diagonal directions. This enables explicit frequency separation. Fig. 4 illustrates the WTconv layer performing a one-level decomposition in the wavelet domain. In the high-frequency subbands, WTConv can specifically extract local edge and texture details, while preserving the global structural information of the image in the low-frequency subband. This multifrequency modeling approach not only avoids the loss of local details but also enhances the model's sensitivity to change region boundaries through the reinforcement of high-frequency features. Algorithm 1 outlines the workflow of the WTConv layer based on WTs, with the goal of retaining high-frequency information.

First, the input image patch $X \in R^{H\times W\times C}$ is convolved with the initial kernel $W^{(0)}$ to extract global low-frequency information $Y_{\mathrm{LL}}^{(0)}$. The kernel size is set to 5×5. Subsequently, the low-frequency component $X_{\mathrm{LL}}^{(i-1)}$ is decomposed via WT into $X_{\mathrm{LL}}^{(i)}$, $X_{\mathrm{LH}}^{(i)}$, $X_{\mathrm{HL}}^{(i)}$, and $X_{\mathrm{HH}}^{(i)}$, representing the low-frequency component and three high-frequency components in different directions (horizontal, vertical, and diagonal). Convolution is then applied to the decomposed low- and high-frequency components using kernels $W^{(i)}$, yielding the convolved features $Y_{\mathrm{LL}}^{(i)}$, $Y_{\mathrm{LH}}^{(i)}$, $Y_{\mathrm{HL}}^{(i)}$, and $Y_{\mathrm{HH}}^{(i)}$. Next, from layer ℓ to layer 1, the inverse WT (IWT) is utilized to fuse the current layer's low- and high-frequency information with the reconstructed feature $Z^{(\ell+1)}$ from the previous layer, reconstructing the feature $Z^{(i)}$. Finally, the initial low-frequency feature $Y_{\mathrm{LL}}^{(0)}$ is added to the reconstruction result $Z^{(1)}$ from the first layer to obtain the output feature map $Z^{(0)}$.

Building on the advantages of WTConv, this study proposes a multilevel decomposition-based WTConv structure, employing WTConv layers with two-level, three-level, four-level, and five-level decompositions to hierarchically extract high-frequency and low-frequency components from the features, comprehensively capturing local details and multiscale information. Specifically, the input features are first decomposed using a two-level Haar WT into one low-frequency feature subband and multiple high-frequency feature subbands, including horizontal, vertical, and diagonal high-frequency components. For each subband, convolutional operations are designed to further extract and enhance frequency-domain features: high-frequency components are processed to capture edge and texture details, while low-frequency components are enhanced to improve global structural representation. Subsequently, the inverse WT is applied to fuse the high-frequency and low-frequency features, reconstructing the complete feature map. To further enhance the extraction of local details, the fused features undergo three-level, four-level, and five-level decomposition and processing in the same manner. At each decomposition level, the aforementioned process is repeated to extract hierarchical detail information. During feature fusion, residual connections are introduced to mitigate potential feature loss caused by wavelet decomposition and reconstruction, ensuring the integrity and effective integration of multilevel information. This hierarchical frequency-aware framework overcomes the limitations of conventional convolution by adaptively aggregating multiscale spectral signatures. Through progressive decomposition, the MLFA module dynamically fuses localized features across frequency bands, enabling precise characterization of both fine-grained textures at varying hierarchical levels.

D. Loss Function

Selecting an appropriate loss function not only accelerates network convergence but also enables the model to extract critical features more effectively. In this study, tailored for hyperspectral CD tasks, we introduce a weighted contrastive loss function to mitigate the impact of class imbalance on model training. This loss function measures the similarity between paired samples and optimizes their distance in the feature space, thereby enhancing the network's ability to differentiate various change types. Furthermore, a weighting mechanism is incorporated into the loss function to balance the contributions of different feature branches to the detection outcomes.

For a given pair of bi-temporal image patches X_1 and X_2 covering the same center pixel (i, j), the feature vectors are extracted by the network as $G(X_1)_{i,j} \in R^C$ and $G(X_2)_{i,j} \in R^C$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

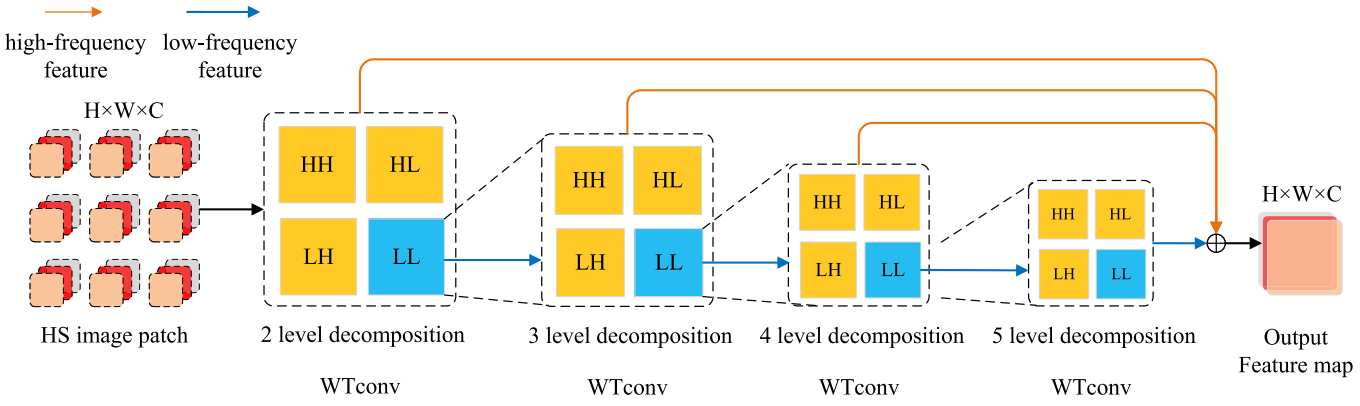


Fig. 3. Structure of the multilevel frequency aware (MLFA) module decomposes hyperspectral features into multiscale frequency subbands via hierarchical WTs, capturing representative local information. By fusing low-frequency (context) and high-frequency (detail) features, it enhances spatial texture representation and sensitivity to subtle local changes, complementing global modeling limitations.

respectively. The Euclidean distance $D(X_1, X2)$ between these two feature vectors is computed as follows:

$$D(X_1, X_2) = \left|\left|G(X_1)_{i,j} - G(X_2)_{i,j}\right|\right|_2. \tag{19}$$

This distance metric is used to measure the similarity between the two pixel pairs in the feature space and is further utilized to optimize the model's loss function. Accordingly, the loss function can be expressed as

$$L(w) = \sum_{k=1}^{P} \left[\left(1 - y_{i,j}^k\right) l_U\left(D_{i,j}^k\right) + y_{i,j}^k l_C\left(D_{i,j}^k\right)\right] \tag{20}$$

where $y_{i,j}^k$ denotes the label of the sample pair, where $y_{i,j}^k = 0$ indicates an unchanged pixel pair and $y_{i,j}^k = 1$ indicates a changed pixel pair. $D_{i,j}^k$ represents the Euclidean distance between the feature vectors of the two patch pairs, $G(X_1)_{i,j}$ and $G(X_2)_{i,j}$. The loss for unchanged pixel pairs $l_U(D_{i,j}^k)$ and the loss for changed pixel pairs $l_C(D_{i,j}^k)$ are defined as follows:

$$l_U\left(D_{i,j}^k\right) = \sigma\left(D_{i,j}^k\right)^2 \tag{21}$$

$$l_C\left(D_{i,j}^k\right) = \max\left(0, q - \sigma\left(D_{i,j}^k\right)\right)^2 \tag{22}$$

where $\sigma(\cdot)$ is the activation function, and q is a hyperparameter used to control the margin between changed pixel pairs. Through this mechanism, the feature vectors of unchanged pixel pairs are encouraged to cluster more closely together, while those of changed pixel pairs are pushed apart, thereby improving the accuracy of CD.

To address the issue of class imbalance, a weighting factor λ is introduced to balance the contributions of the wavelet local branch and the global Mamba branch to the total loss. The final loss function is expressed as

$$\text{Loss}_{\text{total}} = \lambda\,\text{Loss}_{\text{local}} + (1 - \lambda)\,\text{Loss}_{\text{global}}. \tag{23}$$

After executing the aforementioned steps, the pixel values are compared with a predefined threshold. Pixels with values exceeding the threshold are labeled as changed pixels, while the remaining pixels are classified as unchanged.

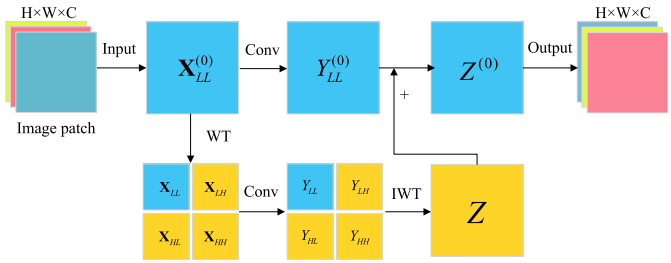


Fig. 4. WTconv layer performs a one-level decomposition in the wavelet domain. The Haar WT decomposes the input into one low-frequency (X_{LL}) and three high-frequency (X_{LH}, X_{HL}, and X_{HH}) components. Subband features are extracted with separate convolutions and fused via inverse WT (IWT), preserving global structure while enhancing local details for improved boundary sensitivity.

IV. Experiments

This section begins with an introduction to three publicly available hyperspectral image CD datasets. Next, we elaborate on the experimental setup, detailing the evaluation criteria and implementation specifics. Finally, we perform various experiments, including hyperparameter tuning, ablation studies, and comparative analyses with state-of-the-art models to thoroughly evaluate the proposed model's performance.

A. Datasets Description

1) River Dataset: The River dataset records changes in a river in Jiangsu Province over different time periods, primarily reflecting river changes due to sediment removal. As shown in Fig. 5, pseudo-color images at two different time points and the corresponding reference maps are provided. This dataset comprises two hyperspectral images acquired by the Hyperion sensor on May 3 and December 31, 2013. The images measure 463 × 241 pixels with a spatial resolution of 30 m. Initially, each image contains 242 spectral bands; after removing noisy bands, 198 bands are retained for the CD task. According to the annotated ground truth, the dataset includes 9698 pixels classified as changed and 101885 as unchanged.

2) Hermiston Dataset: The Hermiston dataset records changes in irrigated farmland in the Hermiston region of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

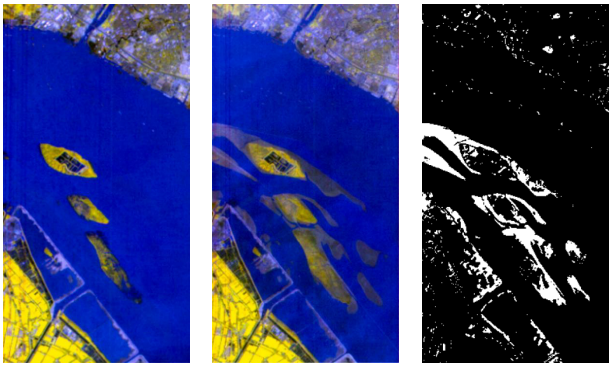


Fig. 5. River dataset. (a) Image acquired on May 3, 2013. (b) Image acquired on December 31, 2013. (c) Ground truth.

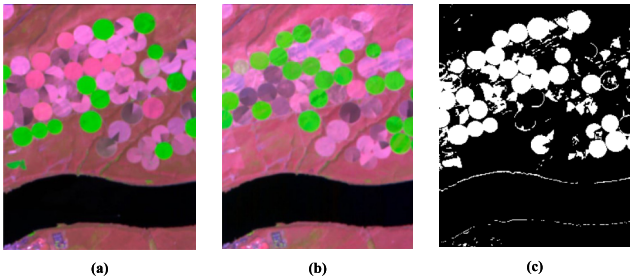


Fig. 6. Hermiston dataset. (a) Image acquired on May 1, 2004. (b) Image acquired on May 8, 2007. (c) Ground truth.

Oregon, USA. As shown in Fig. 6, pseudo-color images at two different time points and the corresponding reference maps are provided. The two hyperspectral images were obtained using the Hyperion sensor aboard the EO-1 satellite on May 1, 2004 and May 8, 2007. Each image has a resolution of 307×241 pixels and contains 154 spectral bands after noise removal. The covered regions include irrigated farmland, rivers, and cultivated areas. This dataset comprises 16676 changed pixels and 57311 unchanged pixels, with changes mainly involving transitions between crops, soil, water bodies, and other surface types.

3) Farmland Dataset: The Farmland dataset reflects changes in farmland near Yancheng, Jiangsu Province, primarily caused by crop rotation. This dataset includes two hyperspectral images acquired by the Hyperion sensor aboard the EO-1 satellite on May 3, 2006 and April 23, 2007. Each image measures 450×140 pixels with a spatial resolution of 30 m. After preprocessing, low signal-to-noise ratio and water absorption bands were removed, leaving 154 spectral bands for subsequent analysis. The annotated ground truth identifies 18277 pixels as changed and 44723 as unchanged, making this dataset well-suited for farmland surface change monitoring studies. Pseudo-color images at the two time points and the corresponding reference maps are shown in Fig. 7.

B. Experimental Configuration

1) Training and Testing Samples Split: For sample division, we randomly selected a specified portion of pixels from each of the three datasets for training purposes. In the River dataset, 2500 unchanged pixels and 1250 changed pixels were randomly sampled, making up 3.36% of the total pixels. For the Farmland dataset, 2101 unchanged and 1049 changed pixels were selected, constituting 5% of the dataset. In the Hermiston dataset, 2441 unchanged pixels and 1258 changed pixels were chosen, also representing 5% of the total samples. Detailed statistics regarding the training and testing sample distributions are presented in Table I.

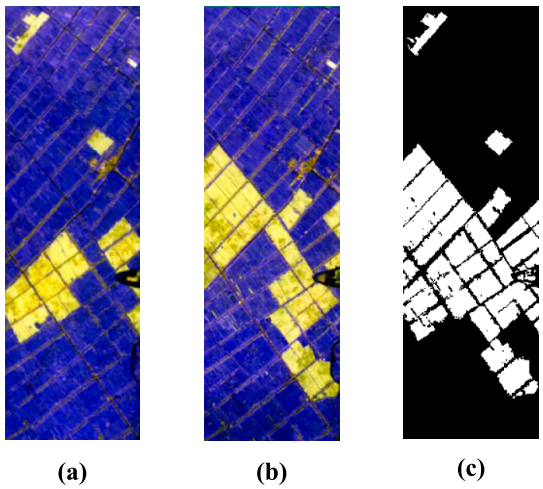


Fig. 7. Farmland dataset. (a) Image acquired on May 3, 2006. (b) Image acquired on April 23, 2007. (c) Ground truth.

TABLE I
NUMBER OF TRAINING AND TEST SAMPLES FOR THREE DIFFERENT DATASETS

Dataset	Training set			Test set		
	Unchanged	changed	Total	Unchanged	changed	Total
Farmland	2101	1049	3150	42622	17228	59850
Hermiston	2441	1258	3699	54870	15418	70288
River	2500	1250	3750	100635	7198	107883

2) Evaluation Metrics: To comprehensively evaluate the performance of the model, this study employs the following commonly used evaluation metrics: overall accuracy (OA), Kappa coefficient (Kappa), and $F1$ score ($F1$). These metrics provide a multidimensional assessment of the model's classification capability and stability.

OA is a key metric that quantifies the proportion of correctly identified pixels relative to the total number of pixels. This metric intuitively reflects the model's classification performance but may be influenced by class imbalance. True positive (TP) denotes the number of correctly classified changed pixels, while true negative (TN) refers to correctly identified unchanged pixels. Conversely, false positive (FP) and false negative (FN) represent the misclassified pixels as changed and unchanged classes, respectively. The formula is expressed as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{24}$$

The Kappa coefficient evaluates classification consistency while accounting for the likelihood of random guessing. It

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

indicates the agreement level between predicted outcomes and the actual ground truth. The observed agreement is denoted as P_o, and P_e signifies the expected agreement under random conditions. The Kappa coefficient varies between [−1, 1], where 1 represents perfect agreement, 0 corresponds to random agreement, and negative values imply worse-than-random consistency. The formula is as follows:

$$\text{Kappa} = \frac{P_o - P_e}{1 - P_e}. \tag{25}$$

$F1$ score represents the harmonic mean of precision and recall, serving as a comprehensive metric for assessing a model's classification performance. By balancing precision and recall, the $F1$ score proves especially valuable in cases involving imbalanced data distributions. The formula is given as follows:

$$F1 = 2 \cdot \frac{\text{Pr} \cdot \text{Pe}}{\text{Pr} + \text{Pe}}. \tag{26}$$

3) Implementation Details: Our method is developed using PyTorch 1.10.1, running on an Intel[1] Core[2] i7-12700F CPU and an NVIDIA GeForce RTX 4090 GPU environment, with Python 3.10.0 as the programming language. The Adam optimizer is employed with a fixed learning rate of 0.0005, without applying a learning rate decay strategy. Based on experimental insights, the training process involves 600 iterations.

C. Comparison Results and Analysis

1) Competitors: To validate the effectiveness of the proposed method, experiments and comparisons are conducted with seven state-of-the-art methods, including CVA, SSMIF, HyperNet, 2D-CNN, GETNET, ML-EDAN, CSANet, and SST-Former. A brief overview of these methods is provided below.

a) CVA [11]: CVA is a classical method for hyperspectral image CD. It detects change areas by comparing the spectral differences between bands. The method is straightforward and serves as a baseline approach for many hyperspectral CD tasks.

b) SSMIF [52]: SSMIF network integrates a spatial–spectral Mamba model, dual-temporal optical flow alignment, and an LSTM module. It leverages SSMs to extract spatiotemporal features from hyperspectral images, utilizes optical flow-based resampling for feature alignment, and employs LSTM to filter redundant information, providing a novel and effective solution for hyperspectral CD.

c) HyperNet [23]: HyperNet is a pixel-level hyperspectral CD framework based on self-supervised learning. It employs a spatial–spectral attention mechanism to separately extract discriminative spatial and spectral features. Furthermore, it introduces an innovative focal cosine similarity loss function, which prioritizes hard-to-distinguish positive samples, enabling robust feature representation without the need for labeled data.

d) 2D-CNN [18]: 2D-CNN utilizes a 2-D convolutional network to extract spatial features and capture spatial correlations through local receptive fields.

e) *GETNET [19]:* GETNET is a general end-to-end 2-D CNN framework for hyperspectral image CD. It introduces a mixed-affinity matrix, effectively mining cross-channel gradient features by integrating subpixel representations and combining spectral and abundance map information.

f) ML-EDAN [24]: The multilevel encoder–decoder attention network adopts a dual-stream encoder–decoder structure and incorporates the context information guided attention module (CIGA) to effectively merge hierarchical spatial–spectral features from multitemporal hyperspectral images. Simultaneously, it utilizes long short-term memory (LSTM) networks to capture temporal dependencies, enhancing both fine-grained details and high-level semantic representations.

g) CSANet [34]: The CSANet extracts joint spatial–spectral–temporal features through a symmetric attention module. This method enhances single-temporal feature representation using a self-attention mechanism and integrates cross-temporal information through a cross-temporal attention module.

h) SST-Former [33]: SST-Former is an end-to-end transformer framework for hyperspectral image CD. It simultaneously models spectral, spatial, and temporal dependencies, overcoming the limitations of traditional convolutional networks by using spectral–spatial transformer and temporal transformer modules to extract long-sequence dependencies and multidimensional feature fusion.

In the prediction results of all methods, white represents the changed areas, while black represents the unchanged areas.

2) Experiment on River: Fig. 8 and Table II present the binary CD maps and quantitative results for the River dataset under different methods. The highest values are highlighted in bold black. As the CVA method is algebraic-based, it is highly sensitive to noise, resulting in a large number of misclassifications in the detection results. The 2D-CNN detects most of the change areas but still struggles with accurately detecting certain smaller regions. GETNET, by introducing a mixed-affinity matrix on top of the 2D-CNN framework, effectively captures subpixel information, performing excellently in change area detection. ML-EDAN strengthens the utilization of multilevel spatial–spectral features through its encoder–decoder structure and attention mechanisms, leading to good performance in both large-scale and fine-grained CD. SSMIF innovatively integrates spatial and spectral SSMs with flow alignment and LSTM filtering, enabling accurate modeling of spatial–spectral dependencies and interactive features, which significantly improves fine-grained CD performance compared to CNN-based methods. HyperNet, through self-supervised training, learns from unlabeled samples, reducing the dependency on labeled data, achieving good results in OA and consistency. However, it is clear that self-supervised learning strategies do not perform as well as other supervised methods. SST-Former enhances sequence representation capabilities by leveraging the global modeling ability of transformer models and the self-attention mechanism, achieving the second-best performance. In comparison, the proposed method in this article provides more accurate CD results than the other methods. By effectively integrating global and local features, it achieves an OA

[1]Registered trademark.
[2]Trademarked.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

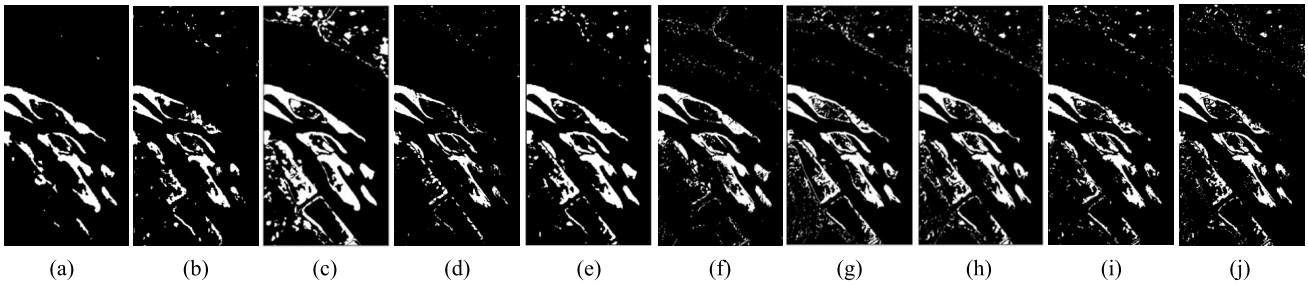


Fig. 8. Experiment result on River dataset. (a) CVA. (b) SSMIF. (c) HyperNet. (d) 2DCNN. (e) GETNET. (f) ML-EDAN. (g) CSANet. (h) SST-Former. (i) Proposed. (j) Ground truth.

TABLE II
QUANTITATIVE CD RESULTS ON RIVER DATASET

	CVA [11]	SSMIF [52]	HyperNet [23]	2DCNN [18]	GETNET [19]	ML-EDAN [24]	CSANet [34]	SST-Former [33]	Proposed
OA(%)	85.73	95.83	94.97	95.17	95.34	95.55	95.76	96.14	**96.65**
Kappa(%)	42.72	75.68	70.95	73.17	74.28	73.51	74.49	77.50	**78.55**

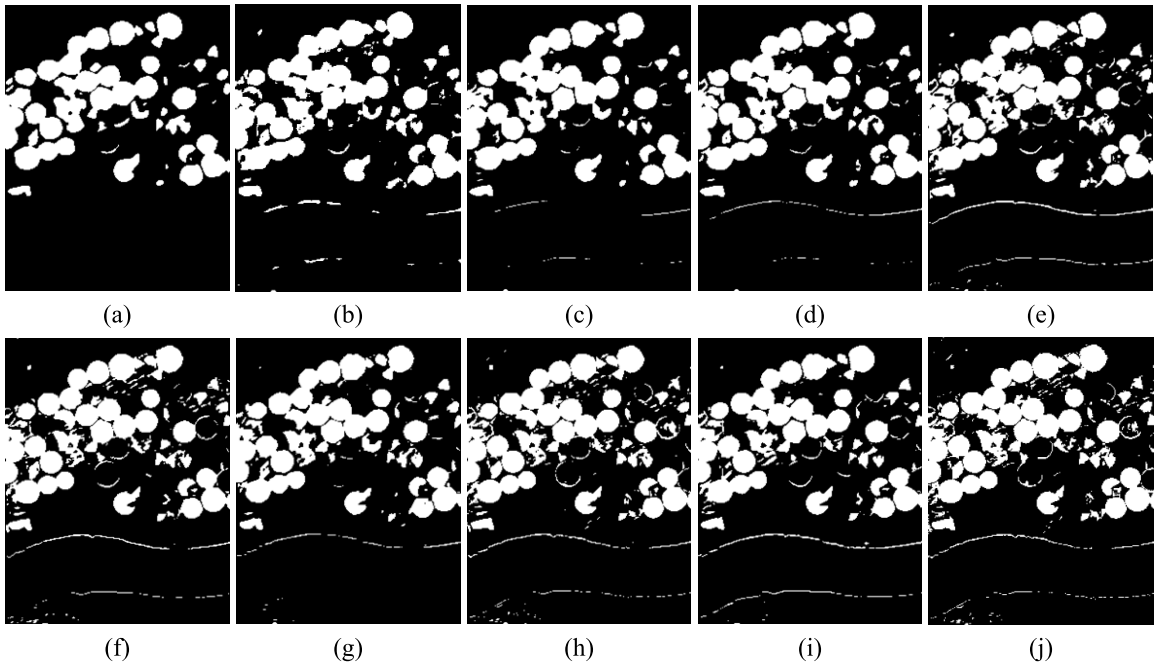


Fig. 9. Experiment result on Hermiston dataset. (a) CVA. (b) SSMIF. (c) HyperNet. (d) 2DCNN. (e) GETNET. (f) ML-EDAN. (g) CSANet. (h) SST-Former. (i) Proposed. (j) Ground truth.

TABLE III
QUANTITATIVE CD RESULTS ON HERMISTON DATASET

	CVA [11]	SSMIF [52]	HyperNet [23]	2DCNN [18]	GETNET [19]	ML-EDAN [24]	CSANet [34]	SST-Former [33]	Proposed
OA(%)	94.54	96.32	93.96	96.00	96.56	96.03	95.92	97.41	**98.94**
Kappa(%)	83.86	86.89	82.21	88.27	90.18	90.21	88.03	91.76	**91.96**

of 96.65% and a Kappa of 78.55%. This demonstrates that the proposed method has significant advantages in accuracy and consistency when distinguishing between changed and unchanged areas, particularly in handling complex scenes such as the River dataset, which consists of dense urban areas and wide rivers.

3) Experiment on Hermiston: Fig. 9 illustrates the CD results on the Hermiston dataset, with the corresponding quantitative evaluations presented in Table III. The changes in the Hermiston dataset are notably intricate, characterized primarily by circular and irregularly curved shapes in the altered regions. Traditional methodologies, such as CVA, exhibit significant

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

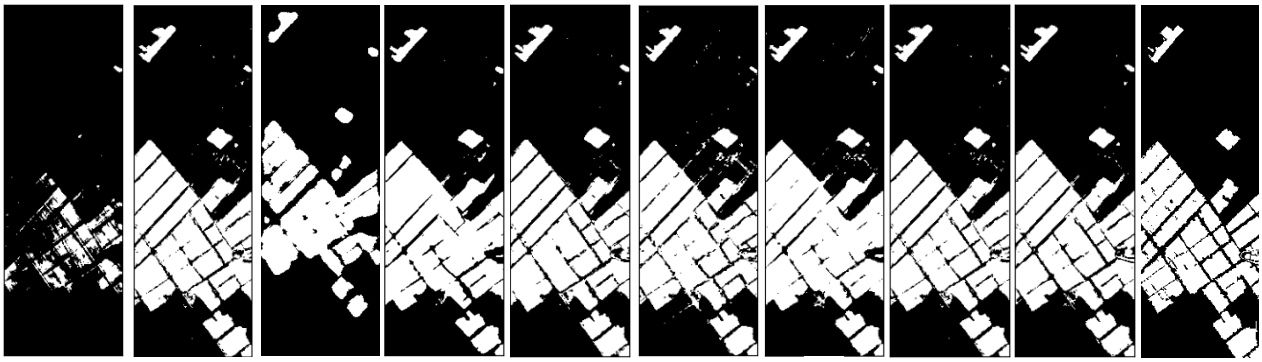


Fig. 10. Experiment result on Farmland dataset. (a) CVA. (b) SSMIF. (c) HyperNet. (d) 2DCNN. (e) GETNET. (f) ML-EDAN. (g) CSANet. (h) SST-Former. (i) Proposed. (j) Ground truth.

TABLE IV
QUANTITATIVE CD RESULTS ON FARMLAND DATASET

	CVA [11]	SSMIF [52]	HyperNet [23]	2DCNN [18]	GETNET [19]	ML-EDAN [24]	CSANet [34]	SST-Former [33]	Proposed
OA(%)	75.23	94.58	90.70	92.54	93.97	93.37	93.82	94.02	**94.95**
Kappa(%)	45.62	87.01	76.83	83.20	86.37	85.06	83.25	85.20	**87.98**

limitations in handling such complex change scenarios, manifesting in substantial omissions and false detections. Although these methods achieve reasonably high OA, their consistency performance remains suboptimal. Furthermore, quantitative analysis reveals that HyperNet yields the lowest metric scores among all evaluated methods. Among the other approaches, 2DCNN demonstrates a detection accuracy with an OA of 96.00% and a Kappa coefficient of 83.27%, slightly inferior to more sophisticated models. GETNET, while effective, introduces salt-and-pepper noise in the detection maps. ML-EDAN and CSANet produce visually clearer detection images; however, these methods often suffer from over-smoothing, leading to inadequate detection capability in fine-grained change regions. SSMIF effectively captures complex spatial patterns via its spatial–spectral Mamba module and further enhances temporal alignment through the bi-temporal flow alignment mechanism, leading to more accurate modeling of irregular change contours. It surpasses CSANet by approximately 0.4% OA. In contrast, SST-Former, leveraging the global modeling capabilities of a transformer, achieves remarkable quantitative performance, with an OA of 97.41% and a Kappa coefficient of 96.76%. Nevertheless, the proposed method in this study further refines the processing of positional information and integrates multiscale high-frequency feature modeling, not only demonstrating superior performance in complex CD scenarios but also achieving higher precision in delineating change region boundaries. Quantitative evaluations indicate that the proposed method attains an OA of 98.94% and a Kappa coefficient of 91.96% on the Hermiston dataset, significantly outperforming all comparative methods. These results underscore the proposed method's exceptional capability in detecting complex change regions and its enhanced robustness and consistency in challenging scenarios.

4) Experiment on Farmland: Fig. 10 presents the CD results on the agricultural dataset. Although the structure of the agricultural dataset is relatively simple, the boundary regions intersecting with ridges are notably complex, thereby increasing the difficulty of CD. Specifically, detection images generated by CVA and 2DCNN algorithms, which rely on raw spectral changes, often exhibit extensive over-detection areas. HyperNet, a self-supervised approach incorporating spatial–spectral attention modules, reduces dependency on annotated data but still fails to deliver satisfactory results in boundary region detection. The introduction of the bi-temporal flow alignment module in SSMIF effectively mitigates the interference of geometric misalignment on detection results and addresses the boundary ambiguity issue that HyperNet fails to resolve. Notably, while ML-EDAN, CSANet, and GETNET achieve comparable OA, CSANet demonstrates the poorest consistency performance among the three. In contrast, the proposed method in this study yields more desirable outcomes, particularly in densely changing central regions, where almost no pixels are misclassified.

Quantitative analysis results for the agricultural dataset are summarized in Table IV, with maximum values highlighted in bold. Specifically, traditional methods such as CVA exhibit relatively low OA and Kappa values, at 75.23% and 45.62%, and 85.99% and 72.43%, respectively. Deep learning-based approaches, including ML-EDAN, CSANet, and GETNET, enhance detection performance through superior feature extraction capabilities, with GETNET achieving an OA of 93.97% and a Kappa coefficient of 86.37%. However, SST-Former and HyperNet still struggle in complex boundary regions, with HyperNet attaining an OA of 90.70% and a Kappa coefficient of 76.83%. The OA of the state-space model-based method SSMIF reached 94.58% and the Kappa coefficient reached 87.01%. In comparison, the pro-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

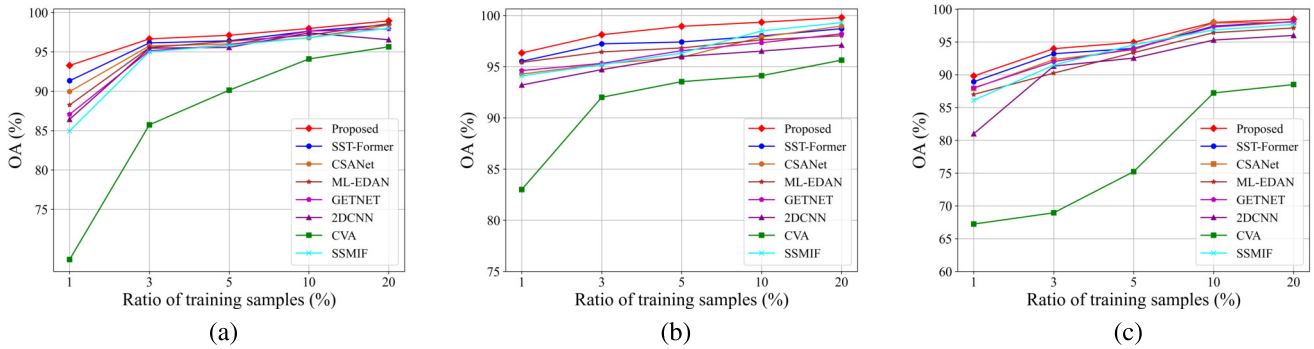


Fig. 11. Effect of training sample ratio on OA across three datasets. (a) River dataset. (b) Hermiston dataset. (c) Farmland dataset.

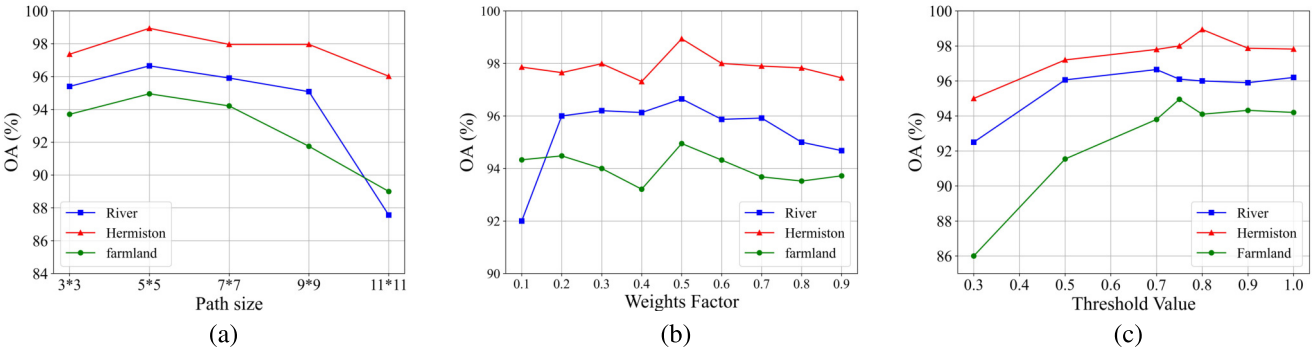


Fig. 12. Different parameter sensitivity analysis in terms of OA on three datasets. (a) results with different patch sizes. (b) Weight factors. (c) Threshold value.

posed method achieves the highest OA of 94.95% and a Kappa coefficient of 87.98%, followed by SSMIF. These results demonstrate that the proposed model, by preserving rich high-frequency information, can extract more detailed sequential change information in fine-grained edge regions, underscoring its superior performance in handling complex boundary scenarios.

D. Effect of Training Sample Ratio

In order to evaluate the effectiveness of the proposed model with different training sample proportions, we conducted five comparative experiments for all supervised learning methods on three datasets: River, Hermiston, and Farmland. The results of the experiments are shown in Fig. 11(a)–(c), respectively. Taking the Hermiston dataset as an example, our method improves the accuracy over the second-best performing method by 0.82%–1.34% across all sample proportions. It is worth noting that the improvement in OA is particularly significant in the high sample proportion range and that our method consistently outperforms the other methods on all datasets.

E. Effect of Hyperparameters

1) Patch Sizes: Fig. 12(a) illustrates the trend of OA with respect to different patch sizes. There is a significant increase in OA as the data patch size increases from 3×3 to 5×5, which indicates that smaller data patch do not adequately capture enough spatial spectral information. However, when the dataset is enlarged to more than 5×5, the OA value decreases significantly, which is attributed to the amplified background noise interference that reduces the discriminability of the features. From the experimental results, the optimal 5×5 patch configuration improves the OA by 0.98%–2.92% compared with the suboptimal settings (3×3, 7×7, and 9×9) for the Hermiston dataset, which achieves the best balance between contextual information acquisition and noise suppression. The experimental results show that a 5×5 patch is always an optimal choice in all three datasets.

2) Weights Factor: We conducted experiments to examine the impact of the weight factor ratio on HSI-CD, selecting values from 0.1 to 0.9 with an interval of 0.1. The results are shown in Fig. 12(b). Specifically, when the weight of the global branch is set to 0.1 and the weight of the local branch is set to 0.9. The experimental results show the best OA on the three datasets when $\lambda = 0.5$, with OA reaching 96.65%, 98.94%, and 94.95% on the River, Hermiston, and Farmland datasets, respectively. This suggests that a balanced distribution of weights between global and local branches positively affects performance. On the contrary, when the weights are overly biased toward one branch, such as $\lambda = 0.1$ or $\lambda = 0.9$, information fusion asymmetry occurs, which negatively affects the model performance. This further reinforces the idea that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

TABLE V
ABLATION EXPERIMENT RESULTS

Methods	River		Hermiston		Farmland	
	OA(%)	Kappa(%)	OA(%)	Kappa(%)	OA(%)	Kappa(%)
Baseline	93.25	31.24	95.99	79.27	92.04	81.11
Without APE	95.93	71.26	97.89	89.87	94.84	87.63
Without MLFA	95.61	70.42	96.93	85.49	94.70	87.32
Proposed	**96.65**	**74.55**	**98.94**	**91.96**	**94.95**	**87.98**

local and global features are equally important in HCI-CD tasks.

3) Threshold Value: For the classification threshold, we conducted experiments with multiple values from 0.1 to 0.9. As shown in Fig. 12(c), the OA of all datasets increases significantly when the threshold is adjusted from 0.3 to 0.7. In particular, the increase in OA is especially obvious in the River dataset, which rises from 86.0% at the threshold value of 0.3 to 93.8% at the threshold value of 0.7. Thereafter, the OA fluctuated slightly, indicating that too high a threshold may weaken the model performance. Based on the results, we determined that the optimal threshold values for the River, Hermiston, and Farmland datasets were 0.8, 0.9, and 0.75, with corresponding OA values of 93.1%, 94.5%, and 92.8%, respectively.

F. Ablation Experiment

To evaluate the effectiveness of the proposed method, we performed ablation studies using three datasets. The baseline model omits the adaptive positional embedding (APE) and MLFA modules. For a thorough assessment, we incrementally introduced these modules and analyzed their influence on OA and Kappa metrics. Comprehensive ablation experiments were conducted on the River, Hermiston, and Farmland datasets to assess each module's role in CD, with the results shown in Table V.

The experimental results indicate that when the network does not include the multilevel frequency-aware module or the adaptive positional embedding, the model's CD performance significantly deteriorates. Specifically, the detection performance on all three datasets is generally better when the adaptive positional embedding module is used, compared to when the multilevel frequency-aware module is omitted, OA improvement of approximately 0.3%–1%. This result suggests that the adaptive positional embedding module plays a more significant role in improving model performance than the multilevel WTconv module, as it effectively captures spatial positional relationships and enhances global modeling capabilities.

More importantly, when both the adaptive positional embedding and the multilevel frequency-aware modules are applied simultaneously, the model's performance is significantly improved, OA increases of approximately 2%–3%, fully validating their complementary roles in local feature extraction and global feature modeling. The synergistic effect of both modules enables effective integration of complex spatial and spectral information, significantly enhancing the model's CD ability. This also demonstrates the proposed network's outstanding capability in leveraging both local details and global contextual information.

G. Computational Complexity

The number of parameters, floating-point operations (FLOPs), training time, and testing time are key indicators for evaluating model efficiency. As illustrated in Figs. 13 and 14, we compare the computational complexity and performance of WDP-Mamba with other state-of-the-art methods on the River, Hermiston, and Farmland datasets.

In terms of parameter size, WDP-Mamba demonstrates a significant advantage. For example, on the River dataset, WDP-Mamba reduces the number of parameters by 36.8% and 19.9% compared to GETNET (154560 k) and SSTFormer (121740 k), respectively. On the Farmland dataset, its parameter count is only 3.86% of that of GETNET. This reduction benefits from the parameter-sharing strategy within the dual-branch architecture. Regarding FLOPs, WDP-Mamba requires 14680 M on the River dataset, which is 83.8% lower than GETNET's 90910 M. Similar reductions are observed on the Hermiston and Farmland datasets, confirming the model's advantage in computational efficiency.

In terms of training and inference efficiency, WDP-Mamba exhibits faster convergence and inference speed than most comparison models. On the River dataset, WDP-Mamba completes training in 3429 s, which is 21.5% faster than GETNET (4369 s), despite having only 63.1% of GETNET's parameters (97540 k). On the Hermiston dataset, the model completes training in 2850 s, a 28.0% reduction compared to GETNET (3960 s), while achieving an OA of 98.94%. On the Farmland dataset, WDP-Mamba achieves 94.95% OA with only 35990 k parameters (just 3.86% of GETNET's), and shortens training time by 6.55% compared to SSTFormer (3050 s). Although in some cases the training time is slightly higher than lightweight models (e.g., on the River dataset, it is 368.4% longer than that of 2DCNN), WDP-Mamba avoids redundant computation through a low-parameter design, achieving an optimal balance between parameter size and computational efficiency. Future research will aim to further reduce computational cost while maintaining or even improving detection performance.

V. DISCUSSION

A. Impact of Different Convolution Methods

In order to evaluate the impact of WTConv compared to other convolution methods on the experimental results,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

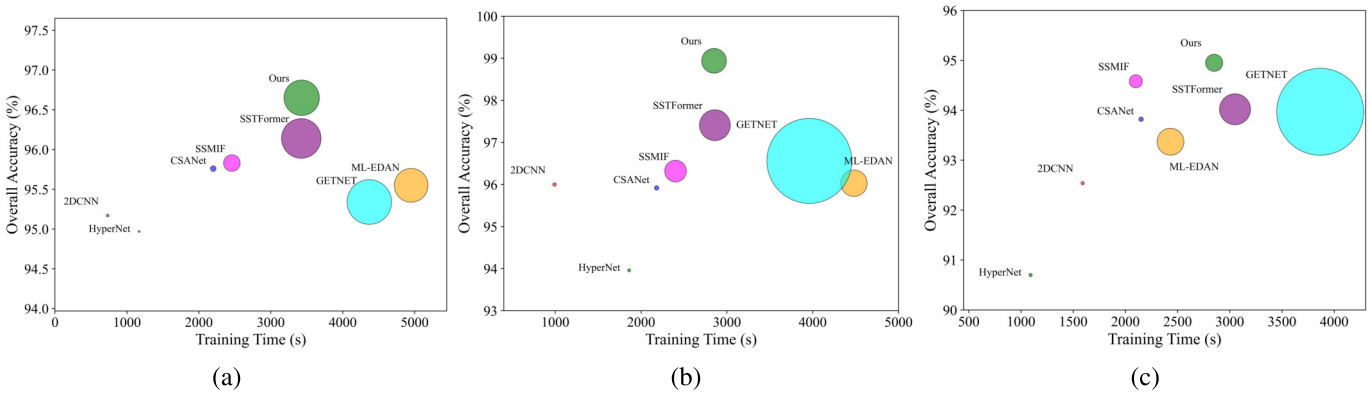


Fig. 13. Comparison of OA, parameter quantities, and training time across three datasets. Larger bubble sizes reflect greater parameter quantities. (a) River. (b) Hermiston. (c) Farmland.

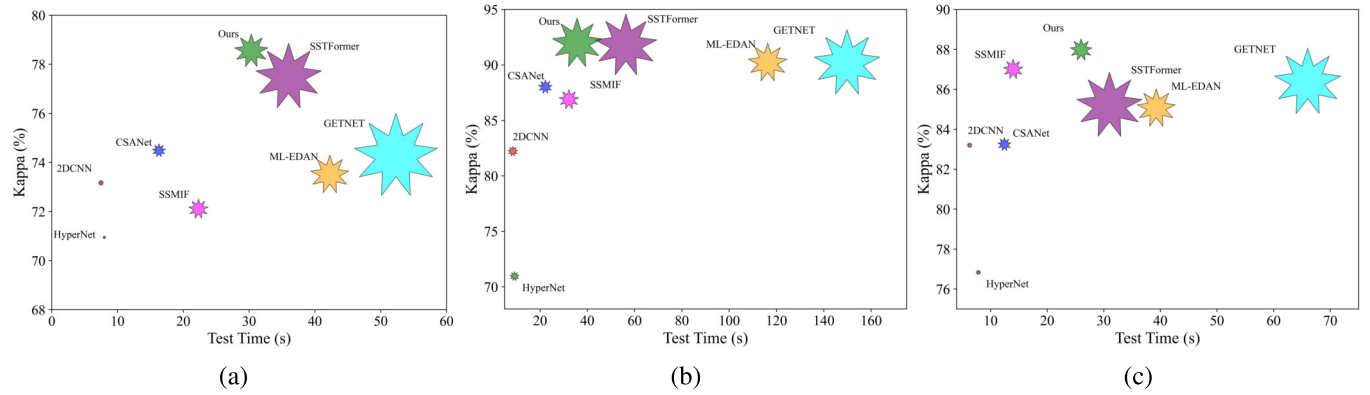


Fig. 14. Comparison of Kappa, FLOPs, and test time across four datasets. Larger starfish sizes reflect higher FLOPs. (a) River. (b) Hermiston. (c) Farmland.

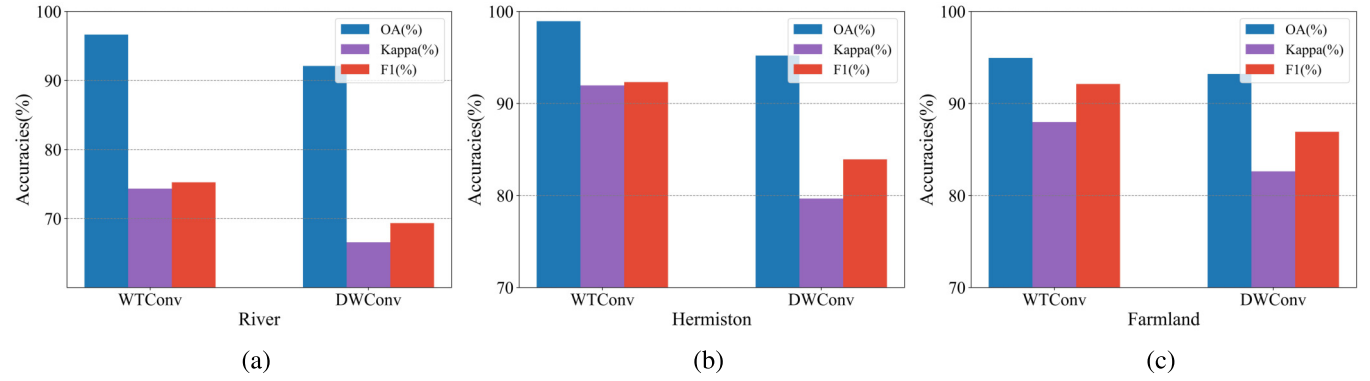


Fig. 15. Impact of different convolutions on the experiment. (a) River dataset. (b) Hermiston dataset. (c) Farmland dataset.

we conducted experiments on three datasets, evaluating the performance of WTConv and depth separable convolution (DWConv) based on OA, Kappa, and F1 metrics. The experimental results are shown in Fig. 15. Specifically, we set the level of WTConv to [2, 3, 4, 5] and replace WTconv with 5×5 depth convolution for comparative analysis. WTConv consistently outperforms DWConv on all three datasets. On the River dataset, WTConv improves 4.54%, 7.77%, and 5.91% compared to DWConv on OA, Kappa, and F1, respectively, and additionally on the Hermiston dataset by 3.73%, 12.29%, and 8.39%, which indicates its outstanding ability in handling scenes with complex spectral and spatial features. On the farmland dataset, the performance of WTConv does not improve significantly compared to DWConv, which may be related to the fact that the spectral information of this dataset is more homogeneous and the boundaries of the feature classes are clearer, which results in the advantage of frequency extraction at different scales not being fully utilized. Overall, WTConv shows excellent performance in the HCI-CD task, especially when analyzing complex scenes.

B. Impact of WTconv With Different Decomposition Levels

To further explore the effect of WTconv decomposition level and kernel size on detection performance, we conducted

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
IMPACT OF WTCONV WITH DIFFERENT DECOMPOSITION LEVELS

Levels	Kernel size	Metric	River	Hermiston	Farmland
[1, 2, 3, 4]	3 × 3	OA	94.46	97.22	93.48
		Kappa	63.54	85.84	84.28
[1, 2, 3, 4]	5 × 5	OA	96.04	97.72	93.21
		Kappa	69.08	88.56	83.29
[1, 2, 3, 4]	7 × 7	OA	96.25	97.99	94.57
		Kappa	73.52	90.03	94.57
[2, 3, 4, 5]	3 × 3	OA	96.17	97.81	94.25
		Kappa	73.02	89.48	86.65
[2, 3, 4, 5]	5 × 5	OA	**96.65**	**98.94**	**94.95**
		Kappa	**74.44**	**91.96**	**87.98**
[2, 3, 4, 5]	7 × 7	OA	96.57	98.41	94.85
		Kappa	73.93	91.48	87.65

TABLE VII
IMPACT OF DIFFERENT WAVELET BASES

Dataset	Harr wavelet			Daubechies wavelet		
	OA(%)	Kappa(%)	F1(%)	OA(%)	Kappa(%)	F1(%)
Farmland	94.95	87.98	92.10	92.05	81.29	87.15
Hermiston	98.94	91.96	92.34	98.59	90.22	91.22
River	96.65	74.44	75.70	92.99	68.25	69.00

experiments on three datasets, with the results presented in Table VI. The experimental results show that increasing the decomposition level and kernel size usually improves the performance. For example, at decomposition level [1, 2, 3, 4], both OA and Kappa tend to increase with increasing kernel size and peak at 7×7 kernels, while at decomposition level [2, 3, 4, 5], the best performance occurs at 5×5 kernels, where the Kappa values of the River, Hermiston, and Farmland datasets reach 74.44%, 91.96% and 87.98% for the River, Hermiston and Farmland datasets, respectively. Meanwhile, under the same kernel size, the decomposition level of [2, 3, 4, 5] always outperforms [1, 2, 3, 4], which indicates that too large a kernel or a lower decomposition level will cause excessive feature aggregation and information redundancy.

C. Impact of Different Wavelet Bases

To verify the rationality of using the Haar wavelet as the primary wavelet basis, we conducted a comparative experiment with Daubechies 4, a widely used representative wavelet from the Daubechies family. As shown in Table VII, although Daubechies 4 performed steadily on the three datasets, it was slightly inferior to Haar in key metrics such as OA, Kappa, and $F1$ score. Specifically, Haar improved OA, Kappa, and $F1$ by 3.66%, 6.19%, and 6.70%, respectively, on the River dataset; similarly, it improved by 2.9%, 6.69%, and 4.95% on the Farmland dataset; and there were also slight improvements on the Hermiston dataset. This is mainly due to the fact that the Haar wavelet uses step-function basis functions with an extremely short filter length, enabling it to sensitively capture abrupt changes and fine-grained structural variations in the image. In contrast, Daubechies 4 has smoother basis functions with longer support and stronger approximation ability, but this characteristic also causes it to blur high-frequency details during processing. In summary, Haar wavelets have greater advantages in CD tasks that rely on edge and detail variations as discriminative features.

VI. CONCLUSION

This article proposes a wavelet-augmented dual-branch position-embedding Mamba network for dense prediction tasks in hyperspectral image analysis. This is the first application of the Mamba method to the HSI-CD task. We innovatively combine the strengths of Mamba and WTConv by designing a dual-branch network structure that accurately captures both global and local features of the image. Building upon this foundation, we introduce the adaptive positional residual state space block in the Mamba network to ensure that spatial and channel positional information is effectively preserved during global modeling, preventing the loss of positional structural information. In addition, we employ a low-parameter multilevel frequency-aware module to decompose the image information into multiple frequency components, thereby enabling the extraction of representative local features across different frequency windows and scales. Finally, by incorporating a weighted contrastive loss function, we assign different weights to the branches to address the balance between local and global information. Experimental results demonstrate that the proposed method performs excellently on the Farmland, Hermiston, and River datasets. Moreover, compared to other CNN or transformer-based methods, it achieves significantly lower parameter counts.

REFERENCES

[1] H. Zhang et al., "BiFA: Remote sensing image change detection with bitemporal feature alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614317.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

[2] J. R. G. Townshend, C. O. Justice, C. Gurney, and J. McManus, "The impact of misregistration on change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 30, no. 5, pp. 1054–1060, Sep. 1992.

[3] C. Persello et al., "Deep learning and Earth observation to support the sustainable development goals: Current approaches, open challenges, and future opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 172–200, Jun. 2022.

[4] D. Chaudhuri and A. Samal, "An automatic bridge detection technique for multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 9, pp. 2720–2727, Sep. 2008.

[5] J. A. Richards and J. A. Richards, *Remote Sensing Digital Image Analysis*, vol. 5. Cham, Switzerland: Springer, 2022.

[6] V. Ferraris, N. Dobigeon, Y. Cavalcanti, T. Oberlin, and M. Chabert, "Coupled dictionary learning for unsupervised change detection between multimodal remote sensing images," *Comput. Vis. Image Understand.*, vol. 189, Dec. 2019, Art. no. 102817.

[7] Y. Fan, Q. Wen, W. Wang, P. Wang, L. Li, and P. Zhang, "Quantifying disaster physical damage using remote sensing data—A technical work flow and case study of the 2014 Ludian earthquake in China," *Int. J. Disaster Risk Sci.*, vol. 8, no. 4, pp. 471–488, Dec. 2017.

[8] J. Yan, L. Wang, W. Song, Y. Chen, X. Chen, and Z. Deng, "A time-series classification approach based on change detection for rapid land cover mapping," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 249–262, Dec. 2019.

[9] R. S. Lunetta and C. D. Elvidge, *Remote Sensing Change Detection: Environmental Monitoring Methods and Applications*. London, U.K.: Taylor & Francis, 1999.

[10] P. R. Coppin and M. E. Bauer, "Digital change detection in forest ecosystems with remote sensing imagery," *Remote Sens. Rev.*, vol. 13, nos. 3–4, pp. 207–234, Apr. 1996.

[11] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2007.

[12] S. Liu, L. Bruzzone, F. Bovolo, M. Zanetti, and P. Du, "Sequential spectral change vector analysis for iteratively discovering and detecting multiple changes in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4363–4378, Aug. 2015.

[13] H. Nemmour and Y. Chibani, "Multiple support vector machines for land cover change detection: An application for mapping urban extensions," *ISPRS J. Photogramm. Remote Sens.*, vol. 61, no. 2, pp. 125–133, Nov. 2006.

[14] X. Li and A. G. O. Yeh, "Principal component analysis of stacked multi-temporal images for the monitoring of rapid urban expansion in the Pearl River Delta," *Int. J. Remote Sens.*, vol. 19, no. 8, pp. 1501–1518, Jan. 1998.

[15] L. M. Bruce, C. H. Koger, and J. Li, "Dimensionality reduction of hyperspectral data using discrete wavelet transform feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 10, pp. 2331–2338, Jan. 2002.

[16] A. A. Nielsen, K. Conradsen, and J. J. Simpson, "Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies," *Remote Sens. Environ.*, vol. 64, no. 1, pp. 1–19, Apr. 1998.

[17] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2858–2874, May 2014.

[18] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[19] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2019.

[20] X. Li, Z. Yuan, and Q. Wang, "Unsupervised deep noise modeling for hyperspectral image change detection," *Remote Sens.*, vol. 11, no. 3, p. 258, Jan. 2019.

[21] Z. Hou, W. Li, L. Li, R. Tao, and Q. Du, "Hyperspectral change detection based on multiple morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022, Art. no. 5507312.

[22] L. Wang, L. Wang, Q. Wang, and P. M. Atkinson, "SSA-SiamNet: Spectral–spatial-wise attention-based Siamese network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022, Art. no. 5510018.

[23] M. Hu, C. Wu, and L. Zhang, "HyperNet: Self-supervised hyperspectral spatial–spectral feature understanding network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5543017.

[24] J. Qu, S. Hou, W. Dong, Y. Li, and W. Xie, "A multilevel encoder–decoder attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 5518113.

[25] Y. Feng, W. Ni, L. Song, and X. Wang, "MsFNet: Multi-scale fusion network based on dynamic spectral features for multi-temporal hyperspectral image change detection," *Remote Sens.*, vol. 16, no. 16, p. 3037, Aug. 2024.

[26] S. Xu et al., "Strengthened residual graph and multiscale gated guided convolutional fusion network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5540014.

[27] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[28] J. Ding, X. Li, J. Li, and S. Chen, "Multiple spatial–spectral features aggregated neural network for hyperspectral change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[29] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.

[30] A. Radford, "Improving language understanding by generative pre-training," 2018.

[31] W. G. C. Bandara and V. M. Patel, "HyperTransformer: A textural and spectral feature fusion transformer for pansharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1767–1777.

[32] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[33] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536814.

[34] R. Song, W. Ni, W. Cheng, and X. Wang, "CSANet: Cross-temporal interaction symmetric attention network for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[35] Z. Chen et al., "TL2GH^{2}T: Triple-path local-to-global network with hybrid head transformer for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528112.

[36] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[37] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[38] T. Huang, X. Pei, S. You, F. Wang, C. Qian, and C. Xu, "LocalMamba: Visual state space model with windowed selective scan," 2024, *arXiv:2403.09338*.

[39] C. Yang et al., "PlainMamba: Improving non-hierarchical mamba in visual recognition," 2024, *arXiv:2403.17695*.

[40] Z. Wang, J.-Q. Zheng, Y. Zhang, G. Cui, and L. Li, "Mamba-UNet: UNet-like pure visual mamba for medical image segmentation," 2024, *arXiv:2402.05079*.

[41] J. Ruan, J. Li, and S. Xiang, "VM-UNet: Vision mamba UNet for medical image segmentation," 2024, *arXiv:2402.02491*.

[42] J. Ma, F. Li, and B. Wang, "U-mamba: Enhancing long-range dependency for biomedical image segmentation," 2024, *arXiv:2401.04722*.

[43] W. Liao, Y. Zhu, X. Wang, C. Pan, Y. Wang, and L. Ma, "LightM-UNet: Mamba assists in lightweight UNet for medical image segmentation," 2024, *arXiv:2403.05246*.

[44] Z. Xing, Y. Tian, Y. Yang, G. Liu, and L. Zhu, "SegMamba: Long-range sequential modeling mamba for 3D medical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*, 2024, pp. 578–588.

[45] J. Zhang, R. Chen, F. Liu, H. Liu, B. Zheng, and C. Hu, "DC-mamba: A novel network for enhanced remote sensing change detection in difficult cases," *Remote Sens.*, vol. 16, no. 22, p. 4186, Nov. 2024.

[46] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[47] H. Wang, Z. Ye, C. Xu, L. Mei, C. Lei, and D. Wang, "TTMGNet: Tree topology mamba-guided network collaborative hierarchical incremental aggregation for change detection," *Remote Sens.*, vol. 16, no. 21, p. 4068, Oct. 2024.

[48] J. N. Paranjape, C. de Melo, and V. M. Patel, "A mamba-based Siamese network for remote sensing change detection," 2024, *arXiv:2407.06839*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.

[49] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Incorporating local clues into mamba for remote sensing image binary change detection," 2024, *arXiv:2406.04207*.
[50] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," 2024, *arXiv:2404.03425*.
[51] Z. Zhang, X. Fan, X. Wang, Y. Qin, and J. Xia, "A novel remote sensing image change detection approach based on multilevel state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4417014.
[52] L. Wu, J. Peng, B. Yang, W. Sun, and Z. Ye, "SSMIF: Enhanced spatial–spectral mamba interactive fusion network for hyperspectral change detection," in *Proc. IEEE Int. Conf. Signal, Inf. Data Process. (ICSIDP)*, Nov. 2024, pp. 1–5.
[53] C. Liu, K. Chen, B. Chen, H. Zhang, Z. Zou, and Z. Shi, "RSCaMa: Remote sensing image change captioning with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[54] S. Zhao, H. Chen, X. Zhang, P. Xiao, L. Bai, and W. Ouyang, "RS-mamba for large remote sensing image dense prediction," 2024, *arXiv:2404.02668*.



Chen Ding (Member, IEEE) received the Ph.D. degree in computer science and technology from Northwestern Polytechnical University, Xi'an, China, in 2018.

He is currently an Associate Professor with the School of Computer Science, Xi'an University of Posts and Telecommunications, Xi'an. In 2019, he presided over the Youth Program of the National Natural Science Foundation of China. His research interests include image processing, machine learning, and pattern recognition.



Xiaofeng Hao received the bachelor's degree from Guangxi University of Science and Technology, Guangxi, China, in 2023, where he is currently pursuing the M.S. degree under the supervision of Dr. Chen Ding.

His research interests include hyperspectral image processing and deep learning.



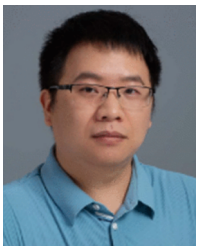
Sirui Zheng received the bachelor's degree from Xi'an University of Posts and Telecommunications, Xi'an, China, in 2024, where he is currently pursuing the M.S. degree under the supervision of Dr. Chen Ding.

His research interests include hyperspectral image processing and deep learning.



Yizhou Dong received the B.S. degree from Chang'an University, Xi'an, China, in 2015, and the Ph.D. degree from Xi'an Jiaotong University, Xi'an, in 2021.

She is currently an Associate Professor at the School of Computer Science, Xi'an University of Posts and Telecommunications. Her research interests include affective computing, depression detection, and artificial intelligence.



Wenqiang Hua (Member, IEEE) received the B.E. degree in electronic science and technology from the University of Electronic Science and Technology of China, Chengdu, China, in 2012, and the Ph.D. degree in circuits and systems from the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University, Xi'an, China, in 2018.

He is currently an Associate Professor with the School of Computer Science and Technology, Xi'an University of Posts and Telecommunications, Xi'an. His research interests include machine learning, deep learning, and PolSAR image processing.



Wei Wei (Senior Member, IEEE) received the Ph.D. degree from Northwestern Polytechnical University, Xi'an, China, in 2012.

He is currently a Professor with the School of Computer Science and Engineering, Northwestern Polytechnical University. He has authored more than 40 articles, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *International Journal of Computer Vision*, and IEEE TRANSACTIONS ON IMAGE PROCESSING. His research interests include image processing, machine learning, and pattern recognition.

Dr. Wei is an Associate Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



Lei Zhang (Member, IEEE) received the Ph.D. degree in computer science and technology from Northwestern Polytechnical University, Xi'an, China, in 2018.

He was a Research Staff with the School of Computer Science, The University of Adelaide, Adelaide, SA, Australia, from 2017 to 2019. He was a Research Scientist with the Inception Institute of Artificial Intelligence (IIAI), Abu Dhabi, United Arab Emirates, from 2019 to 2020. He is currently a Professor with the School of Computer Science, Northwestern Polytechnical University. His research interests include image processing, machine learning, and video analysis.



Yanning Zhang (Fellow, IEEE) received the B.S. degree from the Dalian University of Technology, Dalian, China, in 1988, the M.S. degree from the School of Electronic Engineering, Northwestern Polytechnical University, Xi'an, China, in 1993, and the Ph.D. degree from the School of Marine Engineering, Northwestern Polytechnical University, in 1996.

She is a Professor with the School of Computer Science, Northwestern Polytechnical University. She is also a Cheung Kong Professor of the Ministry of Education, Beijing, China. She has authored more than 200 articles. Her research interests include remote sensing image analysis, computer vision, and pattern recognition.

Dr. Zhang is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and *Pattern Recognition*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:33:54 UTC from IEEE Xplore. Restrictions apply.